Queueing Systems (2020) 96:133–152
https://doi.org/10.1007/s11134-020-09663-x

Analysis of a discrete-time two-class randomly alternating service model with Bernoulli arrivals

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Received: 28 November 2019 / Revised: 29 June 2020 / Published online: 1 August 2020

Abstract

We analyze a discrete-time two-class queueing system with a single server which is alternately available for only one customer class. The server is each time allocated to a customer class for a geometrically distributed amount of time. Service times of the customers are deterministically equal to 1 time slot each. During each time slot, both classes can have at most one arrival. The bivariate process of the number of customers of both classes can be considered as a two-dimensional nearest-neighbor random walk. The generating function of this random walk has to be obtained from a functional equation. This type of functional equation is known to be difficult to solve. In this paper, we obtain closed-form expressions for the joint probability distribution for the number of customers of both classes, in steady state.

Keywords Two-class queueing model · Processor sharing · Singularity analysis · Analytic continuation · Nearest-neighbor random walk

Mathematics Subject Classification 68M20 · 90B22

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1 Introduction

Multi-class queueing systems are characterized by the presence of different classes of customers who are requesting service from the same server(s). Well-known examples of such queuing systems are priority and polling systems [16,17]. Analysis of multi-class queueing systems is often more difficult than the analysis of single queues. The main reason is that in order to describe the system completely and to calculate most performance measures, the joint probability distribution of the number of customers of the different classes in the queueing systems must be found.

In the present paper, we study a two-class queueing model where the server serves customers of either class alternately according to a Bernoulli process. Figure 1 depicts the queueing model that we consider. We assume that the server has no information about the queue contents. As a consequence, when a class with an empty queue is chosen to be served in a particular time slot, no one gets service. This non-work-conserving system is therefore different, but still closely connected, to a discrete-time generalized processor sharing (GPS) queueing model [18]. It has to be noticed that, in contrast to the joint distribution, the marginal distributions of the number of customers present in both queues in this model can be obtained in a relatively easy way, since, from the perspective of one queue, this is a single-sever single-class queueing system with server interruptions [3,8]. In the literature, such systems are sometimes also referred to as systems with server vacations or server breakdowns [9]. Queueing models with server interruptions have been applied very successfully in the past to analyze single-queue contents in queueing systems where customers share a common server, see, for example [13,15] and [19, Sect. 5].

Specifically, this paper considers a discrete-time queueing model with two infinite waiting rooms and one server. As in all discrete-time models, the time axis is divided into fixed-length intervals, referred to as (time) slots in the sequel. New customers may enter the system at any given (continuous) point on the time axis, but services are synchronized to (i.e., can only start and end at) slot boundaries. We further assume that the service of each customer requires exactly one slot. The single server is alternately responsible for the service of two customer classes. This feature is modeled by a single parameter α, which is defined as

$$\alpha := \Pr[\text{server is allocated to class 1 customers during a slot}].$$

We assume that the allocation of the server to a customer class during a slot is independent from slot to slot.

The two input streams of customers into the queueing system are described by means of two independent Bernoulli processes, i.e., the number of class j arrivals

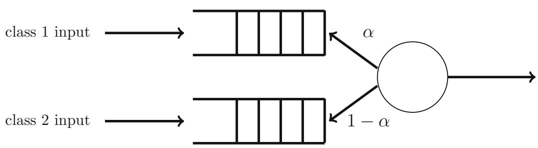

Fig. 1 The considered two-class queueing model

during the consecutive slots is assumed to be Bernoulli-distributed with parameter λ_j, $j = 1, 2$. The common probability generating function is denoted by $A_j(z)$ and is given by

$$A_j(z) = 1 - \lambda_j + \lambda_j z, \quad j = 1, 2. \tag{1}$$

The stability condition is naturally given by the stability conditions of the individual queueing systems, yielding the constraints $\lambda_1 < \alpha < 1 - \lambda_2$.

The objective of this paper is to measure the dependency between the two queues, i.e., obtain the joint probability distribution of the two system contents. As will be shown later, the joint probability generation function of the number of customers of both classes has to be obtained as a solution of a functional equation. The difficulty is that this functional equation contains two unknown partial probability generating functions. In theory, the determination of these functions can be reduced to a Riemann-type boundary value problem [4,10,11,14]. However, this solution requires a conformal mapping which we cannot obtain explicitly. Hence, an additional numerical analysis would be required with the boundary value approach. Other solution methods than the boundary value approach are discussed in [2]. We are primarily interested in obtaining exact results, i.e., without numerical effort. Up till now, only a few classes of two-dimensional random walks are amenable for an exact analysis (for example, [5,6,12]). In [5,6], a nearest-neighbor random walk, stemming from a queueing model, is analyzed. Using analytic continuation, the author showed that the unknown partial generating functions turn out to be meromorphic functions with infinitely many poles, which can be computed by an iterative procedure. In [12], the symmetric shortest queue problem is studied. The approach used in [12] is similar to the one in [5,6], except that in [12] a uniformization technique is used first.

A different approach for analyzing nearest-neighbor random walks (originating from queueing models) is the compensation approach [1]. In this approach, product-form solutions for the balance equations in the interior of the state space are sought. Linear combinations of these product form solutions are constructed, such that the balance equations on the boundaries are also satisfied. This construction can be done iteratively and leads to an infinite series of product forms. It has to be noticed that the compensation approach does not allow transitions from interior states to the North, Northeast and East. Examples of queueing models that satisfy these conditions are the ones studied in [5,6,12]. Since we assume Bernoulli arrivals for both customer classes, the associated two-dimensional random walk corresponding to the system occupancies is a nearest-neighbor random walk as well. However, we emphasize that in our model, one-step transitions to the North and East are possible, which makes the analysis different than those of [5,6,12].

We were able to solve the functional equation, solely with the use of analytic continuation. This approach avoids the explicit calculation of a conformal mapping and yields a nice closed-form expression. As we will show, the two unknown partial generating functions are rational functions with two poles each. It is worth mentioning that, in contrast to the analysis in [6], the *kernel* of the functional equation has two branch points (w.r.t. one variable) outside the unit disk. One would therefore expect that the unknown functions have branch points as well [11]. However, by obtaining a second functional equation we successfully prove that this is not the case.

The outline of this paper is as follows: In the next section, a functional equation for the joint probability generating function of the number of class 1 and class 2 customers is obtained. In Sect. 3, we briefly discuss the marginal distributions and their region of convergence. Properties of the kernel are investigated in Sect. 4. The mathematical focus of this paper is outlined in Sect. 5, where we prove, using analytic continuation, that the unknown partial generating functions are rational functions. In Sect. 6, we provide explicit expressions for the joint probability distribution. In Sect. 7, we study the optimal control of the service allocation when considering the maximum system content. Finally, some conclusions are drawn in Sect. 8.

2 The functional equation

The system contents of class 1 and class 2 customers at the beginning of slot k are denoted by $u_{1,k}$ and $u_{2,k}$, respectively. Further, we denote the number of class 1 and class 2 customers arriving in the system during slot k by $a_{1,k}$ and $a_{2,k}$, respectively. Finally, we denote by r_k the number of available servers for class 1 customers during slot k (0 or 1). The sequence $\{a_{j,k}\}$ constitutes a sequence of i.i.d. Bernoulli random variables with parameter λ_j, $j = 1, 2$. Class 1 and class 2 arrivals are independent, i.e., $a_{1,k}$ and $a_{2,k}$ are independent, for every k. The sequence $\{r_k\}$ constitutes a sequence of i.i.d. Bernoulli random variables with parameter α.

The evolution of the system content from slot k to slot $k + 1$ is described by the following system equations:

$$u_{1,k+1} = (u_{1,k} - r_k)^+ + a_{1,k}, \tag{2}$$

$$u_{2,k+1} = (u_{2,k} - 1 + r_k)^+ + a_{2,k}, \tag{3}$$

where $(\cdot)^+ = \max(\cdot, 0)$. From the system equations, we obtain a relation for the joint probability generating function (pgf) $U_{k+1}(z_1, z_2)$ of the number of type 1 and type 2 customers at the beginning of slot $k + 1$ and the joint pgf $U_k(z_1, z_2)$ of the number of type 1 and type 2 customers at the beginning of slot k:

$$\begin{aligned} U_{k+1}(z_1, z_2) &:= \mathrm{E}\left[z_1^{u_{1,k+1}} z_2^{u_{2,k+1}}\right] \\ &= \frac{A_1(z_1)A_2(z_2)}{z_1 z_2} \{[(1-\alpha)z_1 + \alpha z_2]U_k(z_1, z_2) \\ &\quad +(1-\alpha)(z_2 - 1)z_1 U_k(z_1, 0) + \alpha(z_1 - 1)z_2 U_k(0, z_2)\}, \end{aligned} \tag{4}$$

where we used that $a_{j,k}$, $u_{j,k}$ and r_k are mutually independent, $j = 1, 2$.

Since we are interested in the joint steady-state distribution of $u_{1,k}$ and $u_{2,k}$, we define $U(z_1, z_2)$ as

$$U(z_1, z_2) := \lim_{k\to\infty} \mathrm{E}\left[z_1^{u_{1,k}} z_2^{u_{2,k}}\right].$$

Finally, applying this definition in Eq. (4) we find the following functional equation for $U(z_1, z_2)$:

$$K(z_1, z_2)U(z_1, z_2) = A_1(z_1)A_2(z_2) \times\big[(1-\alpha)(z_2-1)z_1U(z_1,0)+\alpha(z_1-1)z_2U(0,z_2)\big], \tag{5}$$

if $\lambda_1 < \alpha < 1 - \lambda_2$ and where we defined

$$K(z_1, z_2) = z_1z_2 - \big[(1-\alpha)z_1 + \alpha z_2\big]A_1(z_1)A_2(z_2). \tag{6}$$

Throughout the rest of this paper, we will use the following notation:

$$K^{(j)}(x, y) := \left.\frac{\partial K(z_1, z_2)}{\partial z_j}\right|_{z_1=x, z_2=y}, \tag{7}$$

$$K^{(ij)}(x, y) := \left.\frac{\partial^2 K(z_1, z_2)}{\partial z_i \partial z_j}\right|_{z_1=x, z_2=y}, \quad i, j = 1, 2. \tag{8}$$

In the sequel, we will refer to the function K as the kernel. Finally, the joint pgf yields

$$U(z_1, z_2) = \frac{A_1(z_1)A_2(z_2)[(1-\alpha)(z_2-1)z_1U(z_1,0)+\alpha(z_1-1)z_2U(0,z_2)]}{z_1z_2 - [(1-\alpha)z_1+\alpha z_2]A_1(z_1)A_2(z_2)}. \tag{9}$$

There are two unknown functions yet to be determined in the right-hand side of (9), namely the functions $U(z_1, 0)$ and $U(0, z_2)$.

3 The marginal pgfs

From Eq. (9), we easily obtain an expression for $U_1(z)$, describing the system content of class 1 customers:

$$\begin{aligned} U_1(z) :=\ & U(z, 1) \\ =\ & \frac{(\alpha-\lambda_1)(1-\lambda_1+\lambda_1 z)}{\alpha(1-\lambda_1)-\lambda_1(1-\alpha)z}, \end{aligned} \tag{10}$$

in which we already used the normalization condition $U_1(1) = 1$ and canceled the common factor $(z-1)$ in numerator and denominator. The reader will recognize this pgf as the pgf of the system content in a discrete-time $Geo/Geo/1$ system with mean arrival rate λ_1 and mean service rate α. The corresponding probability mass function (pmf) is given by

$$\Pr[u_1 = 0] = 1 - \frac{\lambda_1}{\alpha},$$
$$\Pr[u_1 = n] = \frac{\alpha - \lambda_1}{\alpha(1-\alpha)} \frac{1}{\tau_1^n}, \quad n \geq 1,$$

where τ_1 is the unique zero of the denominator in (10), given by

$$\tau_1 = \frac{\alpha}{1-\alpha} \frac{1-\lambda_1}{\lambda_1}. \tag{11}$$

We can calculate the pgf $U_2(z)$ describing the system content of class 2 customers in a similar way:

$$\begin{aligned} U_2(z) &:= U(1, z) \\ &= \frac{(1-\alpha-\lambda_2)(1-\lambda_2+\lambda_2 z)}{(1-\alpha)(1-\lambda_2) - \alpha\lambda_2 z}. \end{aligned} \tag{12}$$

This is the pgf of the number of customers in a discrete-time $Geo/Geo/1$ system with mean arrival rate λ_2 and mean service rate $1-\alpha$. The corresponding pmf is given by

$$\Pr[u_2 = 0] = 1 - \frac{\lambda_2}{1-\alpha},$$
$$\Pr[u_2 = n] = \frac{1-\alpha-\lambda_2}{\alpha(1-\alpha)} \frac{1}{\tau_2^n}, \quad n \geq 1,$$

where we defined

$$\tau_2 = \frac{1-\alpha}{\alpha} \frac{1-\lambda_2}{\lambda_2}. \tag{13}$$

Finally, we consider the marginal pgf $U_T(z)$ of the total number of customers. This pgf is given by

$$\begin{aligned} U_T(z) &:= U(z, z) \\ &= \frac{(1-\lambda_1+\lambda_1 z)(1-\lambda_2+\lambda_2 z)}{(1-\lambda_1)(1-\lambda_2) - \lambda_1\lambda_2 z}[(1-\alpha)U(z, 0) + \alpha U(0, z)]. \end{aligned} \tag{14}$$

The dominant singularity of $U_T(z)$ is either the dominant singularity of $U(z, 0)$, the dominant singularity of $U(0, z)$ or the zero of the denominator τ_T, defined by

$$\tau_T = \frac{(1-\lambda_1)(1-\lambda_2)}{\lambda_1\lambda_2} = \tau_1\tau_2. \tag{15}$$

Notice that the unknown functions $U(z, 0)$ and $U(0, z)$ are in fact partial pgfs, because they equal the following power series:

$$U(z,0)=\sum_{n=0}^{\infty}\Pr[u_1=n,u_2=0]z^n, \tag{16}$$

$$U(0,z)=\sum_{n=0}^{\infty}\Pr[u_1=0,u_2=n]z^n. \tag{17}$$

Because $\Pr[u_1=n,u_2=0]\le\Pr[u_1=n]$, $\forall n\ge 0$, the power series (16) converges at least for $|z|<\tau_1$. Hence, $U(z,0)$ has to be analytic in $|z|<\tau_1$. Similarly, $U(0,z)$ is analytic in $|z|<\tau_2$.

By determining the singularities of (the analytically continued) $U(z,0)$ and $U(0,z)$, we will obtain explicit expressions for these two functions.

4 Analysis of the kernel $K(z_1,z_2)$

Let us define two regions in $\mathbb{C}^2$:

$$\Omega_1=\{(z_1,z_2)\mid |z_1|<\tau_1,|z_2|\le 1\}, \tag{18}$$

$$\Omega_2=\{(z_1,z_2)\mid |z_1|\le 1,|z_2|<\tau_2\}. \tag{19}$$

Now notice that for $(z_1,z_2)\in\Omega_1\cup\Omega_2$, the double power series expansion of $U(z_1,z_2)$, i.e.,

$$\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\Pr[u_1=n,u_2=m]z_1^n z_2^m,$$

converges and it is consequently finite in this region. For a zero (z_1,z_2) of the kernel

$$K(z_1,z_2)=0,\quad (z_1,z_2)\in\Omega_1\cup\Omega_2,$$

it is seen that Eq. (5) implies that

$$(1-\alpha)(z_2-1)z_1U(z_1,0)+\alpha(z_1-1)z_2U(0,z_2)=0.$$

It is clear that the zeros of the function K play an important role in order to obtain conditions for the unknown functions $U(z,0)$ and $U(0,z)$. In the remainder of this section, we will study the zeros of $K(z_1,z_2)$ as a function of z_2 for fixed z_1 and vice versa.

4.1 The function $Y_1(z)$

We rewrite $K(z_1, z_2)$ as follows:

$$K(z_1, z_2) = -\alpha\lambda_2 A_1(z_1)z_2^2 + (z_1 - A_1(z_1)[\alpha(1-\lambda_2) + (1-\alpha)\lambda_2 z_1])z_2 \\ - (1-\alpha)(1-\lambda_2)z_1 A_1(z_1). \tag{20}$$

Considering $K(z_1, z_2)$ as a polynomial of degree 2 in the variable z_2, the corresponding discriminant is given by

$$D_Y(z_1) = (z_1 - A_1(z_1)[\alpha(1-\lambda_2) + (1-\alpha)\lambda_2 z_1])^2 - 4\alpha(1-\alpha)\lambda_2(1-\lambda_2)z_1 A_1^2(z_1).$$

Let us define the zeros of the polynomial $D_Y(z)$ by x_1, x_2, x_3 and x_4.

Lemma 1 *The zeros of $D_Y(z)$ are real, moreover*

$$0 < x_1 < x_2 < 1 < \tau_T < x_3 < x_4 < \infty.$$

Proof We define

$$h(x) := x - A_1(x)[\alpha(1-\lambda_2) + (1-\alpha)\lambda_2 x],$$

which is a polynomial of degree 2.

One can easily verify that $h(0) < 0$, $h(1) > 0$ and $\lim_{x\to\infty} h(x) < 0$. Hence, $h(x)$ must have a zero in $]0, 1[$, and in $]1, \infty[$. Moreover, we have that the latter zero is larger than τ_T because $h(\tau_T) = \tau_T(\alpha - 2\alpha\lambda_1 + \lambda_1) > 0$. It now easily follows that D_Y is strictly negative in the two zeros of $h(x)$. Since $D_Y(0) > 0$, $D_Y(1) > 0$, $D_Y(\tau_T) > 0$ and $\lim_{x\to\infty} D_Y(x) > 0$, the zeros of $D_Y(z)$ are real and we have necessarily that $0 < x_1 < x_2 < 1 < \tau_T < x_3 < x_4 < \infty$. □

The function $D_Y(z)$ can thus be factorized as

$$D_Y(z) = (1-\alpha)^2\lambda_1^2\lambda_2^2(z - x_1)(z - x_2)(z - x_3)(z - x_4).$$

Further, we define

$$\Delta(z) := (1-\alpha)\lambda_1\lambda_2\sqrt{z - x_1}\sqrt{z - x_2}\sqrt{z - x_3}\sqrt{z - x_4}, \tag{21}$$

with $\sqrt{\cdot}$ the principal value of the square root. Next, we define the following two functions:

$$Y_1(z) = \frac{z - A_1(z)[\alpha(1-\lambda_2) + (1-\alpha)\lambda_2 z] + \Delta(z)}{2\alpha\lambda_2 A_1(z)}, \tag{22}$$

$$Y_2(z) = \frac{z - A_1(z)[\alpha(1-\lambda_2) + (1-\alpha)\lambda_2 z] - \Delta(z)}{2\alpha\lambda_2 A_1(z)}. \tag{23}$$

One easily sees that $K(z, Y_1(z)) = 0$, $K(z, Y_2(z)) = 0$, and

$$Y_1(z)Y_2(z) = \tau_2 z, \quad \forall z \in \mathbb{C}. \tag{24}$$

The denominators in (22) and (23) have one unique zero, namely $-\frac{1-\lambda_1}{\lambda_1}$. The numerator in (22) vanishes for $-\frac{1-\lambda_1}{\lambda_1}$. Therefore, $Y_1(z)$ is analytic in $\mathbb{C} \setminus [x_1, x_2] \cup [x_3, x_4]$. Further, we have that

$$\lim_{z\to\infty} \left| \frac{Y_1(z)}{Y_2(z)} \right| = 0,$$

and $|Y_1(z)| = |Y_2(z)|$ for $z \in [x_1, x_2] \cup [x_3, x_4]$. Applying the maximum modulus principle to the function $\frac{Y_1(z)}{Y_2(z)}$ yields

$$|Y_1(z)| < |Y_2(z)|, \quad z \in \mathbb{C} \setminus [x_1, x_2] \cup [x_3, x_4]. \tag{25}$$

4.2 The function $X_1(z)$

We now consider $K(z_1, z_2)$ as a polynomial in the variable z_1. The discriminant is again a polynomial of degree 4, given by

$$D_X(z_2) = (z_2 - A_2(z_2)[(1-\alpha)(1-\lambda_1) + \alpha\lambda_1 z_2])^2 - 4\alpha(1-\alpha)\lambda_1(1-\lambda_1)z_2 A_2(z_2)^2.$$

The zeros of $D_X(z)$ are denoted by y_1, y_2, y_3 and y_4.

Lemma 2 *The zeros of $D_X(z)$ are real, moreover*

$$0 < y_1 < y_2 < 1 < \tau_T < y_3 < y_4 < \infty.$$

Proof Analogously to Lemma 1. □

Let us define

$$\Psi(z) := \alpha\lambda_1\lambda_2\sqrt{z-y_1}\sqrt{z-y_2}\sqrt{z-y_3}\sqrt{z-y_4}, \tag{26}$$

and two functions

$$X_1(z) = \frac{z - A_2(z)[(1-\alpha)(1-\lambda_1) + \alpha\lambda_1 z] + \Psi(z)}{2(1-\alpha)\lambda_1 A_2(z)}, \tag{27}$$

$$X_2(z) = \frac{z - A_2(z)[(1-\alpha)(1-\lambda_1) + \alpha\lambda_1 z] - \Psi(z)}{2(1-\alpha)\lambda_1 A_2(z)}. \tag{28}$$

It can be seen that $K(X_1(z), z) = 0$, $K(X_2(z), z) = 0$ and

$$X_1(z)X_2(z) = \tau_1 z, \quad \forall z \in \mathbb{C}. \tag{29}$$

One can then again verify that $X_1(z)$ is an analytic function for $z \in \mathbb{C} \setminus [y_1, y_2] \cup [y_3, y_4]$. Finally, using the maximum modulus principle, one can show that

$$|X_1(z)| < |X_2(z)|, \quad z \in \mathbb{C} \setminus [y_1, y_2] \cup [y_3, y_4]. \tag{30}$$

5 Analytic continuation of $U(z, 0)$ and $U(0, z)$

We analytically continue the functions $U(z, 0)$ and $U(0, z)$. Such a continuation is unique, and the extended functions restricted to $|z| < \tau_1$, $|z| < \tau_2$, coincide with the power series expansions (16) and (17). We will denote the analytic continuation also by $U(z, 0)$ and $U(0, z)$. The final result of this procedure will be that the (analytically continued) functions are rational functions with only two simple poles.

5.1 Continuation to $|z| < \tau_T$

As mentioned in Sect. 4, a zero of the kernel $K(z_1, z_2)$, $(z_1, z_2) \in \Omega_1 \cup \Omega_2$, yields a condition for $U(z_1, 0)$ and $U(0, z_2)$. First, we need to investigate the modulus of the functions $X_1(z)$ and $Y_1(z)$ to be able to obtain such conditions.

Lemma 3 1. *If* $1 < |z| < \tau_T$, *then* $|X_1(z)| < |z|$. *Moreover, if* $1 < |z| < \tau_2$, *we have that* $|X_1(z)| < 1$.
2. *If* $1 < |z| < \tau_T$, *then* $|Y_1(z)| < |z|$. *Moreover, if* $1 < |z| < \tau_1$, *we have that* $|Y_1(z)| < 1$.

Proof We shall only prove the first statement, the second one can be proven analogously.

We first prove that for every z_2, $1 < |z_2| < \tau_T$, there is a unique $z_1 =: X(z_2)$ with $|X(z_2)| < |z_2|$, such that $K(X(z_2), z_2) = 0$.

Let z_2 be a fixed complex number with modulus r such that $1 < r < \tau_T$. For $|z_1| = r$, we have $|z_1 z_2| = r^2$ and

$$\begin{aligned}|((1-\alpha)z_1 + \alpha z_2)A_1(z_1)A_2(z_2)| &\le ((1-\alpha)|z_1| + \alpha|z_2|)A_1(|z_1|)A_2(|z_2|)\\ &= rA_1(r)A_2(r)\ .\end{aligned}$$

Consider the equation $f(r) = r - A_1(r)A_2(r)$. Since $f(1) = 0$, $f'(1) > 0$, and τ_T is the smallest root of f which is strictly greater than one, we have that $r > A_1(r)A_2(r)$ for $r \in]1, \tau_T[$. Hence, we have the following inequality:

$$|((1-\alpha)z_1 + \alpha z_2)A_1(z_1)A_2(z_2)| \le rA_1(r)A_2(r) < r^2 = |z_1 z_2|.$$

Application of Rouché's theorem using the contour $|z_1| = r$ yields the existence of a unique $X(z_2)$, with $|X(z_2)| < r$, such that $K(X(z_2), z_2) = 0$.

Next, let z_2 be a fixed complex number with modulus r such that $1 < r < \tau_2$. For $|z_1| = 1$, we have $|z_1 z_2| = r$ and

$$
\begin{aligned}
|((1-\alpha)z_1+\alpha z_2)A_1(z_1)A_2(z_2)| &\leq ((1-\alpha)|z_1|+\alpha|z_2|)A_1(|z_1|)A_2(|z_2|)\\
&= (1-\alpha+\alpha r)A_2(r).
\end{aligned}
$$

We remark that now $r \in]1, \tau_2[$. By a similar argument as above, we must have $((1-\alpha)+\alpha r)A_2(r) < r$. Hence, we have the inequality

$$
|((1-\alpha)z_1+\alpha z_2)A_1(z_1)A_2(z_2)| \leq (1-\alpha+\alpha r)A_2(r) < r = |z_1 z_2|.
$$

By Rouché's theorem applied to the contour $|z_1| = 1$ we can conclude that, for fixed z_2, $1 < |z_2| < \tau_2$, $K(z_1, z_2)$ has exactly one zero inside the unit circle.

Because $|X_1(z)| < |X_2(z)|$, we necessarily have that $X(z) = X_1(z)$. Therefore, the lemma is proven. □

Without loss of generality, we assume $\tau_1 \leq \tau_2$ for the remainder of this section. We now want to refine the result of Lemma 3 in the region $\tau_1 \leq |z| < \tau_T$. Let us define the function $g(x)$, for real x, such that $A_1(g(x))A_2(g(x)) = x$ and $g(1) = 1$. An explicit expression for $g(x)$ is given by

$$
g(x) = \frac{2\lambda_1\lambda_2 - \lambda_1 - \lambda_2 + \sqrt{4\lambda_1\lambda_2 x + (\lambda_1-\lambda_2)^2}}{2\lambda_1\lambda_2}. \tag{31}
$$

By the definition of τ_T, 1 and τ_T are the only fix-points of $A_1(x)A_2(x)$ in $[1, \tau_T]$. Therefore, 1 and τ_T are the only fix-points of $g(x)$ in $[1, \tau_T]$. It follows that τ_T is an attractive fix-point of $g(x)$, because $g(\tau_T) = \tau_T$ and $g'(\tau_T) = (A'(\tau_T))^{-1} < 1$, while 1 is not an attractive fix-point of $g(x)$, since $g(1) = 1$ and $g'(1) = (\lambda_1+\lambda_2)^{-1} > 1$.

We now define the sequence $\delta_0 = \tau_1$, $\delta_{n+1} = g(\delta_n)$, $\forall n \in \mathbb{N}$. It follows that $\delta_n \to \tau_T$, for $n \to \infty$. We now have the following refinement of Lemma 3.

Lemma 4 $|Y_1(z)| < \delta_n$ *and* $|X_1(z)| < \delta_n$ *if* $\delta_n \leq |z| < \delta_{n+1}$, $\forall n \in \mathbb{N}$.

Proof Let z_2 be fixed with modulus r such that $\delta_n \leq r < \delta_{n+1}$. For $|z_1| = \delta_n$, we have that $|z_1 z_2| = r\delta_n$ and

$$
\begin{aligned}
|((1-\alpha)z_1+\alpha z_2)A_1(z_1)A_2(z_2)| &\leq ((1-\alpha)|z_1|+\alpha|z_2|)A_1(|z_1|)A_2(|z_2|)\\
&\leq ((1-\alpha)\delta_n+\alpha r)A_1(\delta_n)A_2(r)\\
&< rA_1(\delta_{n+1})A_2(\delta_{n+1})\\
&= r\delta_n.
\end{aligned}
$$

By Rouché's theorem applied to the contour $|z_1| = \delta_n$, we have that for fixed z_2, $\delta_n \leq |z_2| < \delta_{n+1}$, there exists a unique z_1 such that $|z_1| < \delta_n$ and $K(z_1, z_2) = 0$. Again, we necessarily have that $z_1 = X_1(z_2)$.

The same proof can be applied to prove that $|Y_1(z)| < \delta_n$ if $\delta_n \leq |z| < \delta_{n+1}$. □

We are now ready to analytically continue $U(z, 0)$ and $U(0, z)$ outside $|z| < \tau_1$ and $|z| < \tau_2$, respectively.

Theorem 1 1. *The functions $U(z,0)$ and $U(0,Y_1(z))$ can be analytically continued in $\tau_1 \le |z| < \tau_T$, $z \ne \tau_1$.*
2. *The functions $U(0,z)$ and $U(X_1(z),0)$ can be analytically continued in $\tau_2 \le |z| < \tau_T$, $z \ne \tau_2$.*

Proof The main idea of the proof is to substitute $(z_1, z_2) = (z, Y_1(z))$ into the functional equation (5), such that LHS vanishes. We then obtain a relation between $U(z,0)$ and $U(0,Y_1(z))$. Similarly, substituting $(z_1,z_2) = (X_1(z), z)$ yields a relation between $U(X_1(z),0)$ and $U(0,z)$. We then make use of these two relations to analytically continue $U(z,0)$ and $U(0,z)$.

Because of Lemma 3, we have that $|Y_1(z)| < 1$, if $1 < |z| < \tau_1$; consequently $U(z, Y_1(z))$ remains finite and we obtain the following relation as explained in Sect. 4:

$$(1-\alpha)(Y_1(z)-1)zU(z,0) = -\alpha(z-1)Y_1(z)U(0,Y_1(z)), \quad 1<|z|<\tau_1, \tag{32}$$

which we rewrite as

$$U(z,0) = -\frac{\alpha(z-1)Y_1(z)U(0,Y_1(z))}{(1-\alpha)z(Y_1(z)-1)}, \qquad 1<|z|<\tau_1. \tag{33}$$

According to Lemma 4, for $\tau_1 \le |z| < \delta_1$, we have that $|Y_1(z)| < \tau_1 \le \tau_2$. Hence, the right-hand side in (33) is analytic in $\tau_1 \le |z| < \delta_1$, $z \ne \tau_1$. We can therefore analytically continue $U(z,0)$ for $\tau_1 \le |z| < \delta_1$, $z \ne \tau_1$ via Eq. (33).

Using again Lemma 3, we obtain

$$(1-\alpha)(z-1)X_1(z)U(X_1(z),0) = -\alpha(X_1(z)-1)zU(0,z), \quad 1<|z|<\tau_2. \tag{34}$$

From Lemma 4, we have that $|X_1(z)| < \tau_1$, if $\tau_1 \le |z| < \delta_1$. Therefore, $U(0,z)$ can be analytically continued for $\tau_1 \le |z| < \delta_1$, $z \ne \tau_2$ via

$$U(0,z) = -\frac{(1-\alpha)(z-1)X_1(z)U(X_1(z),0)}{\alpha z(X_1(z)-1)}. \tag{35}$$

We exclude $z=\tau_2$, because $X_1(\tau_2)=1$ and the right-hand side of (35) is consequently not analytic in τ_2.

Now suppose that $U(z,0)$ is analytic for $|z|<\delta_n$, $z\ne\tau_1$ and $U(0,z)$ is analytic for $|z|<\delta_n$, $z\ne\tau_2$ for a certain $n\in\mathbb{N}$. According to Lemma 4, we have that $|Y_1(z)|<\delta_n$, for $\delta_n \le |z| < \delta_{n+1}$. Consequently, the right-hand side of (33) is analytic in $1<|z|<\delta_{n+1}$, $z\ne\tau_1$. It follows that $U(z,0)$ can be analytically continued into $\delta_n \le |z| < \delta_{n+1}$. Similarly, because we have that $|X_1(z)|<\delta_n$, for $\delta_n \le |z| < \delta_{n+1}$, we can analytically continue $U(0,z)$ into $\delta_n \le |z| < \delta_{n+1}$ via Eq. (35). By induction, we have shown that $U(z,0)$ is analytic for $|z|<\delta_n$, $z\ne\tau_1$, $\forall n$ and that $U(0,z)$ is analytic for $|z|<\delta_n$, $z\ne\tau_2$, $\forall n$. Since $\delta_n\to\tau_T$ for $n\to\infty$ it follows that $U(z,0)$ is analytic in $|z|<\tau_T$, $z\ne\tau_1$, and that $U(z,0)$ is analytic in $|z|<\tau_T$, $z\ne\tau_2$. □

From Theorem 1, it follows that τ_1 is an isolated singularity of $U(z,0)$. Multiplying Eq. (33) by $(z-\tau_1)$ and taking the limit to τ_1 yields

$$\lim_{z\to\tau_1}(z-\tau_1)U(z,0) = -\frac{\alpha-\lambda_1}{(1-\alpha)^2}\frac{1-\lambda_1-\lambda_2}{\lambda_1}. \tag{36}$$

Because the above limit is strictly negative (and hence different from zero), τ_1 is a simple pole of $U(z,0)$. Obviously, τ_1 must be the radius of convergence of the power series (16). Analogously, it follows that τ_2 is the radius of convergence of the power series (17) and that τ_2 is a simple pole of $U(0,z)$ with residue

$$\lim_{z\to\tau_2}(z-\tau_2)U(0,z) = -\frac{1-\alpha-\lambda_2}{\alpha^2}\frac{1-\lambda_1-\lambda_2}{\lambda_2}. \tag{37}$$

5.2 Continuation to $|z| \geq \tau_T$

5.2.1 The composition $Y_1(X_1(Y_1(z)))$

We start this section with a useful observation for values where $Y_1'(z)$ and $X_1'(z)$ vanish. Notice that we can compute $Y_1'(z)$ as

$$Y_1'(z) = -\frac{K^{(1)}(z,Y_1(z))}{K^{(2)}(z,Y_1(z))}. \tag{38}$$

Consider now a value z^* for which $Y_1'(z^*) = 0$. From Eq. (38), it follows that $K^{(1)}(z^*,Y_1(z^*)) = 0$. Hence, z^* is a zero with multiplicity (at least) two of the polynomial $z \mapsto K(z,Y_1(z^*))$. This is only possible if $D_X(Y_1(z^*)) = 0$. According to Lemma 2, we must have that $Y_1(z^*)$ is equal to either y_1, y_2, y_3 or y_4. Further, we cannot have that also $Y_1''(z^*) = 0$, because otherwise $K^{(11)}(z,Y(z^*)) = 0$, which is impossible because $D_X(z)$ has no zeros with multiplicity three or more.

We now study the graph of $Y_1(z)$ on the real interval $[1,\tau_1]$. Note that $Y_1(z)$ and $Y_1'(z)$ are continuous on $[1,\tau_1]$. It can be shown that $Y_1(1) = 1$, $Y_1(\tau_1) = 1$, $Y_1(\tau_T) = \tau_2$.[1] The derivative of $Y_1(z)$, evaluated at $z=1$, $z=\tau_1$ and $z=\tau_T$, is given by

$$Y_1'(1) = -\frac{\alpha-\lambda_1}{1-\alpha-\lambda_2} < 0, \tag{39}$$

$$Y_1'(\tau_1) = \frac{(1-\alpha)\lambda_1(\alpha-\lambda_1)}{\alpha(1-\lambda_1)(1-\lambda_1-\lambda_2)} > 0, \tag{40}$$

and

$$Y_1'(\tau_T) = \frac{(1-\alpha)\lambda_1(1-\lambda_1-\lambda_2)}{\alpha(\alpha-\lambda_1)(1-\lambda_1)} > 0, \tag{41}$$

respectively.

[1] We remark that $\Delta(z) = -\sqrt{D_Y(z)}$ if $z \in [1,\tau_T]$, due to the properties of the principal square root.

From (39) and (40), it follows that there exists a $v^* \in]1, \tau_1[$ such that $Y_1'(v^*) = 0$. Because of Lemma 2 and the fact that $|Y_1(z)| < 1$ for $1 < |z| < \tau_1$, there can be at most two values such that the derivative of $Y_1'(z)$ in $]1, \tau_T[$ vanishes. Hence, v^* is the only value in $]1, \tau_1[$ such that $Y'(z)$ vanishes. Similarly, there cannot exist a value in the interval $]\tau_1, \tau_T[$ such that $Y_1'(z)$ vanishes, since $|Y_1(z)| < |z|$ for $\tau_1 \leq |z| < \tau_T$, $Y_1'(\tau_1) > 0$, $Y_1(\tau_1) = 1$ and because of Lemma 2. We conclude that $Y_1(z)$ is strictly decreasing for $z \in [1, v^*[$ and strictly increasing for $z \in]v^*, \tau_T]$. Analogously, there exists a $w^* \in]1, \tau_2[$ such that $X_1(z)$ is strictly decreasing for $z \in [1, w^*[$ and strictly increasing for $z \in]w^*, \tau_T]$.

For every z, either $X_1(Y_1(z)) = z$ or $X_2(Y_1(z)) = z$, by the definitions of $X_1(z)$ and $X_2(z)$, for example, for $z = 1$, we have that $X_1(Y_1(1)) = 1$, while $X_2(Y_1(1)) = \tau_1$. Starting in $z = 1$, we use the (real) inverse function theorem to investigate in what region the equation $X_1(Y_1(z)) = z$ holds. It can be seen that the inverse function theorem will work as long as $Y_1(z)$ is continuously differentiable and $Y_1'(z) \neq 0$. Because of the analysis above, this yields $X_1(Y_1(s)) = s$ if $s \in [1, v^*[$ and $X_2(Y_1(s)) = s$ if $s \in]v^*, \tau_T]$. From Eq. (29), we get

$$X_1(Y_1(s)) = \tau_1 \frac{Y_1(s)}{s}, \quad s \in]v^*, \tau_T] \; .$$

Analogously, we have that

$$Y_1(X_1(s)) = \tau_2 \frac{X_1(s)}{s}, \quad s \in]w^*, \tau_T].$$

Substituting $s = Y_1(s)$ into the above equation yields

$$Y_1(X_1(Y_1(s))) = \tau_2 \frac{X_1(Y_1(s))}{Y_1(s)}, \quad Y_1(s) \in]w^*, \tau_T].$$

Because $Y_1(\tau_1) = 1$ and $Y_1(\tau_T) = \tau_2$, there exists a $t^* \in]\tau_1, \tau_T[$ such that $Y_1(t^*) = w^*$. Hence, $Y_1(s) > w^*$ if $s \in]t^*, \tau_T]$, because $Y_1(s)$ is strictly increasing on $[t^*, \tau_T]$. Because also $v^* < \tau_1 \leq \tau_2 < t^*$, we have the following key property:

$$Y_1(X_1(Y_1(s))) = \frac{\tau_T}{s}, \quad s \in]t^*, \tau_T]. \tag{42}$$

5.2.2 A new functional equation

From the proof in Theorem 1, we have that, for $|z| < \tau_T$,

$$(1-\alpha)\frac{zU(z,0)}{z-1} = -\alpha \frac{Y_1(z)U(0, Y_1(z))}{Y_1(z)-1}, \tag{43}$$

$$\alpha \frac{zU(0,z)}{z-1} = -(1-\alpha)\frac{X_1(z)U(X_1(z),0)}{X_1(z)-1}. \tag{44}$$

If we restrict z to the real interval $]t^*, \tau_t[$ and use consecutively Eqs. (43), (44) and (42), we find that

$$\begin{aligned}(1-\alpha)\frac{zU(z,0)}{z-1} &= -\alpha\frac{Y_1(z)U(0,Y_1(z))}{Y_1(z)-1}\\ &= (1-\alpha)\frac{X_1(Y_1(z))U(X_1(Y_1(z)),0)}{X_1(Y_1(z))-1}\\ &= -\alpha\frac{Y_1(X_1(Y_1(z)))U(0,Y_1(X_1(Y_1(z))))}{Y_1(X_1(Y_1(z)))-1}\\ &= -\alpha\tau_T\frac{U(0,\frac{\tau_T}{z})}{\tau_T - z}, \quad z\in]t^*,\tau_T[, \end{aligned} \tag{45}$$

which can be rewritten as

$$U(z,0) = -\frac{\alpha\tau_T(z-1)U(0,\frac{\tau_T}{z})}{(1-\alpha)z(\tau_T - z)}, \quad z\in]t^*,\tau_T[. \tag{46}$$

Since $U(0,\frac{\tau_T}{z})$ is analytic for $|z|>1$, $z\neq\tau_1$, the right-hand side of (45) is analytic for $|z|>1$, $z\neq\{\tau_1,\tau_T\}$. Hence, the equality is valid for $1<|z|<\tau_T$, $z\neq\tau_1$ and $U(z,0)$ can be analytically continued for $|z|\geq\tau_T$, $z\neq\tau_T$ via Eq. (46). $U(z,0)$ is therefore analytic in $\mathbb{C}\setminus\{\tau_1,\tau_T\}$. Using (46), it follows that

$$\lim_{z\to\tau_T}(z-\tau_T)U(z,0) = \frac{(\alpha-\lambda_1)(1-\lambda_1-\lambda_2)}{(1-\alpha)\lambda_1\lambda_2}. \tag{47}$$

Hence, τ_T is a simple pole of $U(z,0)$.

Further, using (45) it follows that $U(0,z)$ is analytic in $\mathbb{C}\setminus\{\tau_2,\tau_T\}$, and τ_T is a simple pole of $U(0,z)$ with residue

$$\lim_{z\to\tau_T}(z-\tau_T)U(0,z) = \frac{(1-\alpha-\lambda_2)(1-\lambda_1-\lambda_2)}{\alpha\lambda_1\lambda_2}. \tag{48}$$

Using the new functional equation (46), it also easily follows that

$$\lim_{z\to\infty}U(z,0) = 0, \tag{49}$$

$$\lim_{z\to\infty}U(0,z) = 0. \tag{50}$$

We are now ready to present our main theorem.

Theorem 2 1. *$U(z,0)$ is a rational function and its partial fraction expansion is given by*

$$U(z,0) = \frac{(\alpha-\lambda_1)(1-\lambda_1-\lambda_2)}{(1-\alpha)} \times \left(\frac{-1}{(1-\alpha)\lambda_1 z - \alpha(1-\lambda_1)} + \frac{1}{\lambda_1\lambda_2 z - (1-\lambda_1)(1-\lambda_2)} \right). \tag{51}$$

2. *$U(0,z)$ is a rational function and its partial fraction expansion is given by*

$$U(0,z) = \frac{(1-\alpha-\lambda_2)(1-\lambda_1-\lambda_2)}{\alpha} \times \left(\frac{-1}{\alpha\lambda_2 z - (1-\alpha)(1-\lambda_2)} + \frac{1}{\lambda_1\lambda_2 z - (1-\lambda_1)(1-\lambda_2)} \right). \tag{52}$$

Proof We will only prove part 1, since part 2 can be obtained analogously.

$U(z,0)$ is an analytic function with the exception of τ_1 and τ_T as simple poles. From (36) and (47), we obtain that the singular part of $U(z,0)$ at τ_1 and τ_T is given by $-\frac{(\alpha-\lambda_1)(1-\lambda_1-\lambda_2)}{(1-\alpha)^2\lambda_1(z-\tau_1)}$ and $\frac{(\alpha-\lambda_1)(1-\lambda_1-\lambda_2)}{(1-\alpha)\lambda_1\lambda_2(z-\tau_T)}$, respectively. Hence,

$$L(z) := U(z,0) - \left(-\frac{(\alpha-\lambda_1)(1-\lambda_1-\lambda_2)}{(1-\alpha)^2\lambda_1(z-\tau_1)} + \frac{(\alpha-\lambda_1)(1-\lambda_1-\lambda_2)}{(1-\alpha)\lambda_1\lambda_2(z-\tau_T)} \right)$$

is an entire function. Because $U(z,0) \to 0$ as $z \to \infty$, we have that $L(z) \to 0$ as $z \to \infty$ as well. Hence, $L(z)$ is bounded and tends to zero (as $z \to \infty$). From Liouville's theorem, we can conclude that $L(z) = 0$. □

6 The joint distribution

Substituting Eqs. (51) and (52) into (9) yields

$$U(z_1,z_2) = \frac{(\alpha-\lambda_1)(1-\alpha-\lambda_2)(1-\lambda_1-\lambda_2)(1-\lambda_1+\lambda_1 z_1)(1-\lambda_2+\lambda_2 z_2)}{(\alpha(1-\lambda_1)-(1-\alpha)\lambda_1 z_1)((1-\alpha)(1-\lambda_2)-\alpha\lambda_2 z_2)} \times \frac{(1-\lambda_1)(1-\lambda_2)-\lambda_1\lambda_2 z_1 z_2}{((1-\lambda_1)(1-\lambda_2)-\lambda_1\lambda_2 z_1)((1-\lambda_1)(1-\lambda_2)-\lambda_1\lambda_2 z_2)}. \tag{53}$$

The joint pgf $U(z_1,z_2)$ is now completely determined in terms of the system parameters α, λ_1 and λ_2. From this joint pgf, we can obtain the joint pmf.

Theorem 3 *The joint probability distribution of class 1 and class 2 customers is given by*

$$\Pr[u_1=0,u_2=0] = \frac{(1-\alpha-\lambda_2)\,(\alpha-\lambda_1)\,(1-\lambda_1-\lambda_2)}{(1-\lambda_2)\,\alpha\,\,(1-\alpha)\,(1-\lambda_1)},$$

$$\Pr[u_1=n,u_2=0] = \frac{(\alpha-\lambda_1)(1-\lambda_1-\lambda_2)}{(1-\alpha)(1-\lambda_1)} \left(\frac{1}{\alpha}\frac{1}{\tau_1^n} - \frac{1}{1-\lambda_2}\frac{1}{\tau_T^n} \right), \quad n \geq 0,$$

$$\begin{aligned}\Pr[u_1 = 0, u_2 = n] =& \frac{(1-\alpha-\lambda_2)(1-\lambda_1-\lambda_2)}{\alpha(1-\lambda_2)} \\ &\times \left(\frac{1}{(1-\alpha)}\frac{1}{\tau_2^n} - \frac{1}{1-\lambda_1}\frac{1}{\tau_T^n}\right), \quad n \geq 0, \\ \Pr[u_1 = n, u_2 = m] =& \frac{(\alpha-\lambda_1)(1-\lambda_1-\lambda_2)}{\alpha(1-\alpha)\lambda_1(1-\lambda_1)}\frac{1}{\tau_1^n\tau_T^m} \\ &+ \frac{(1-\alpha-\lambda_2)(1-\lambda_1-\lambda_2)}{\alpha(1-\alpha)\lambda_2(1-\lambda_2)}\frac{1}{\tau_2^m\tau_T^n} \\ &- \frac{(1-\lambda_1-\lambda_2)^2}{\lambda_1(1-\lambda_1)\lambda_2(1-\lambda_2)}\frac{1}{\tau_T^{n+m}}, \quad n \geq 1, m \geq 1. \end{aligned} \tag{54}$$

Proof Expanding the factors in the denominator of (53) with respect to z_1 and z_2 yields the result. It can be verified that $\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\Pr[u_1 = n, u_2 = m] = 1$ and that $\Pr[u_1 = n, u_2 = m] \geq 0, \forall n, m \in \mathbb{N}$. □

Several performance measures can be obtained from the joint pgf (53). As an example, we give the covariance between the number of class 1 and class 2 customers:

$$\mathrm{cov}[u_1, u_2] = -\frac{\lambda_1\lambda_2(1-\lambda_1)(1-\lambda_2)}{(1-\lambda_1-\lambda_2)^2}. \tag{55}$$

Expression (55) gives two interesting results. First, the covariance is independent of the system parameter α, which is, in our opinion, a surprising result. Notice that the correlation between both system contents does depend on the parameter α, because the variances of u_1 and u_2 depend on α. Secondly, the covariance is always negative, which could be expected because if u_1 is exceptionally large, then either there were a lot of type 1 arrivals in the previous time slots, or type 1 customers were not served often during the last couple of slots. In the latter case, it is likely that u_2 is small.

7 Optimal control of the alternating service policy

In practice, designing concrete systems, it is natural to determine the parameter α such that a certain cost function is minimized. Consider, for example, the expected maximum system content $\mathrm{E}[\max(u_1, u_2)]$ as the cost function for our queueing system, in which α can be considered as the decision variable. We want to compute the optimal value of α, denoted by α^*, that minimizes the cost function, i.e., we want to compute

$$\alpha^* = \arg\min_{\alpha} \mathrm{E}[\max(u_1, u_2)], \quad \lambda_1 < \alpha < 1 - \lambda_2. \tag{56}$$

Using the joint pmf (54) of Theorem 3 and the fact that $\Pr[\max(u_1, u_2) > L] = 1 - \Pr[u_1 \leq L, u_2 \leq L]$, the expected maximum system content is given by

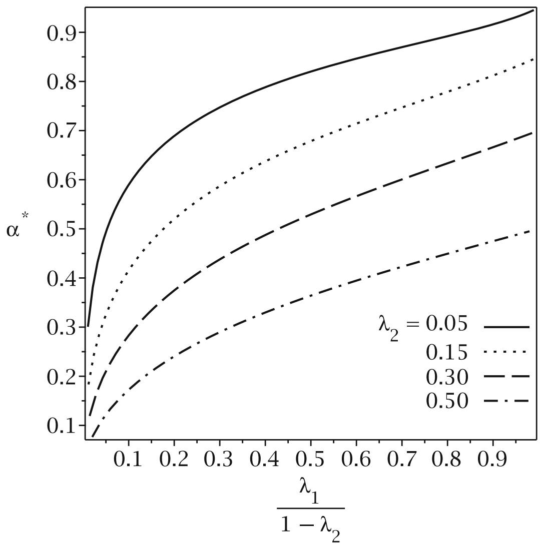

Fig. 2 The optimal value α^* that minimizes the expected maximum system content $\mathrm{E}[\max(u_1, u_2)]$ versus the class 1 arrival rate λ_1, for various values of λ_2

$$
\begin{aligned}
\mathrm{E}[\max(u_1, u_2)] &= \sum_{L=0}^{\infty} \Pr[\max(u_1, u_2) > L] \\
&= \frac{\tau_T}{\tau_T^2 - 1} + \frac{\lambda_1(1-\lambda_1)}{\alpha - \lambda_1} - \frac{(1-\lambda_1)(1-\lambda_2)}{\alpha(1-\lambda_1)\tau_T - (1-\alpha)\lambda_1} \\
&\quad + \frac{\lambda_2(1-\lambda_2)}{1-\alpha-\lambda_2} - \frac{(1-\lambda_1)(1-\lambda_2)}{(1-\alpha)(1-\lambda_2)\tau_T - \alpha\lambda_2}.
\end{aligned} \tag{57}
$$

$\mathrm{E}[\max(u_1, u_2)]$ is, as a function in α, a rational function with poles λ_1, $1-\lambda_2$, $\frac{\lambda_1}{(1-\lambda_1)\tau_T+\lambda_1}$ and $\frac{(1-\lambda_2)\tau_T}{\lambda_2+(1-\lambda_2)\tau_T}$. Using the fact that $\tau_T > 1$, it follows that $\frac{\lambda_1}{(1-\lambda_1)\tau_T+\lambda_1} < \lambda_1$ and $1-\lambda_2 < \frac{(1-\lambda_2)\tau_T}{\lambda_2+(1-\lambda_2)\tau_T}$ so that $\mathrm{E}[\max(u_1, u_2)]$ is well defined for $\alpha \in]\lambda_1, 1-\lambda_2[$. Moreover, $\mathrm{E}[\max(u_1, u_2)]$ is strictly positive and convex for $\alpha \in]\lambda_1, 1-\lambda_2[$. Hence, we have that $\mathrm{E}[\max(u_1, u_2)]$ has a unique minimizer $\alpha^* \in]\lambda_1, 1-\lambda_2[$. Therefore, the minimizer α^* of (57) can easily be computed using numerical software. Figure 2 illustrates the optimal values of α as a function of λ_1, for $\lambda_2 = 0.05, 0.15, 0.3$ and 0.5. Because the stability conditions imply that $\lambda_1 + \lambda_2 < 1$, we scaled the horizontal axis by dividing by $1-\lambda_2$, so that the four curves have the same domain $[0, 1]$. From Fig. 2, it can be seen that the optimal α, as a function of λ_1, increases the most for λ_1 close to 0. When $\lambda_1 \to 1-\lambda_2$, the optimal α approaches $1-\lambda_2$. This is due to the stability conditions $\lambda_1 < \alpha < 1-\lambda_2$.

8 Conclusion

In this paper, we analyzed a discrete-time two-class queueing system with Bernoulli arrivals where a single server is alternately responsible for the service of the two customer classes. Analytic continuation was used to obtain the joint pgf of the system contents. Further, we have obtained closed-form expressions for the joint pmf of the system contents. Multiple future directions of this work are possible. A first possibility is the extension to general arrival processes. Our conjecture is that here, however, restrictions also have to be made, for example, arrival processes with rational pgfs. Recently, we have obtained a numerical approximation for this class of arrival processes, based on the dominant singularities of the unknown generating functions [7]. Secondly, one could consider a three-class model. Since hardly any results are known for multi-class queues with more than two classes, we expect that this extension will be the most difficult. We here briefly sketch possible approaches for the three-class case. In analogy with this paper, let us assume that the three-dimensional joint PGF of the system contents is denoted by $U(z_1, z_2, z_3)$. Three unknown functions are then to be determined, namely $U(0, z_2, z_3)$, $U(z_1, 0, z_3)$ and $U(z_1, z_2, 0)$. An idea would be to first focus on the joint distribution of two queues, for example $U(z_1, z_2, 1)$. However, the new kernel K for the fundamental functional equation of the joint PGF $U(z_1, z_2, 1)$ of these two queues is of a different nature than the kernel we encountered in this paper. Yet another idea is to merge two queues and analyze the joint distribution between this merged queue and the third queue, for example by focusing on $U(z, z, z_3)$. However, because two customers can arrive in the merged queue within a time slot, the arrival process to this merged queue is no longer Bernoulli. Therefore, this idea brings us back to the analysis of more general arrival processes. Finally, as a third possible extension, the assumption that the allocation of the server to a customer class is independent from slot to slot could be dropped and replaced with a more general assumption.

References

1. Adan, I.J.B.F., Wessels, J., Zijm, W.H.M.: A compensation approach for two-dimensional Markov processes. Adv. Appl. Probab. **25**(4), 783–817 (1993)
2. Adan, I.J., Boxma, O.J., Resing, J.A.C.: Queueing models with multiple waiting lines. Queueing Syst. **37**(1–3), 65–98 (2001)
3. Bruneel, H., Kim, B.G.: Discrete-Time Models for Communication Systems Including ATM, vol. 205. Springer, Berlin (2012)
4. Cohen, J.W., Boxma, O.J.: Boundary Value Problems in Queueing System Analysis. North-Holland, Amsterdam (1983)
5. Cohen, J.W.: On the Determination of the Stationary Distribution of a Symmetric Clocked Buffered Switch. Teletraffic Science and Engineering, vol. 2, pp. 297–307. Elsevier, Amsterdam (1997)
6. Cohen, J.W.: On the asymmetric clocked buffered switch. Queueing Syst. **30**(3–4), 385–404 (1998)
7. Devos, A., Fiems, D., Walraevens, J., Bruneel, H.: An approximate analysis of a Bernoulli alternating service model. In: Phung-Duc, T., Kasahara, S., Wittevrongel, S. (eds.) Queueing Theory and Network Applications. QTNA 2019. Lecture Notes in Computer Science, vol. 11688. Springer, Cham (2019)
8. Fiems, D., Steyaert, B., Bruneel, H.: Randomly interrupted GI-G-1 queues: service strategies and stability issues. Ann. Oper. Res. **112**(1–4), 171–183 (2002)
9. Fiems, D., Maertens, T., Bruneel, H.: Queueing systems with different types of server interruptions. Eur. J. Oper. Res. **188**(3), 838–845 (2008)
10. Fayolle, G., Iasnogorodski, R.: Two coupled processors: the reduction to a Riemann–Hilbert problem. Z. Wahrscheinlichkeitstheorie verwandte Gebiete **47**(3), 325–351 (1979)
11. Fayolle, G., Malyshev, V.A., Iasnogorodski, R.: Random Walks in the Quarter-Plane. Springer, Berlin (1999)
12. Kingman, J.F.: Two similar queues in parallel. Ann. Math. Stat. **32**(4), 1314–1323 (1961)

13. LaMaire, R.O.: An M/G/1 vacation model of an FDDI station. IEEE J. Sel. Areas Commun. **9**(2), 257–264 (1991)
14. Malyshev, V.A.: An analytical method in the theory of two-dimensional positive random walks. Sib. Math. J. **13**(6), 917–929 (1972)
15. Núñez-Queija, R.: Sojourn times in a processor sharing queue with service interruptions. Queueing Syst. **34**(1–4), 351–386 (2000)
16. Takagi, H.: Queueing Analysis: A Foundation of Performance Evaluation Volume 1: Vacation and Priority Systems, Part 1. North-Holland, Amsterdam (1991)
17. Takagi, H.: Queuing analysis of polling models. ACM Comput. Surv. (CSUR) **20**, 5–28 (1998)
18. Vanlerberghe, J., Maertens, T., Walraevens, J., De Vuyst, S., Bruneel, H.: On the optimization of two-class work-conserving parameterized scheduling policies. 4OR-Q J. Oper. Res. **14**, 281–308 (2016)
19. Walraevens, J., Steyaert, B., Bruneel, H.: Performance analysis of a single-server ATM queue with a priority scheduling. Comput. Oper. Res. **30**(12), 1807–1829 (2003)